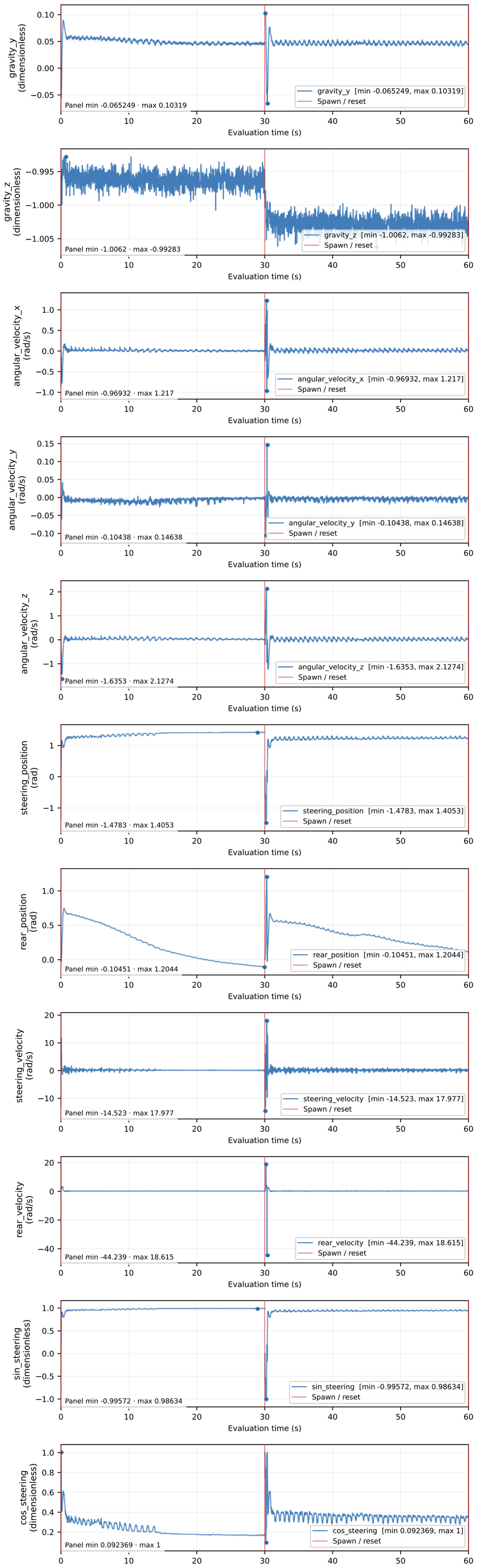
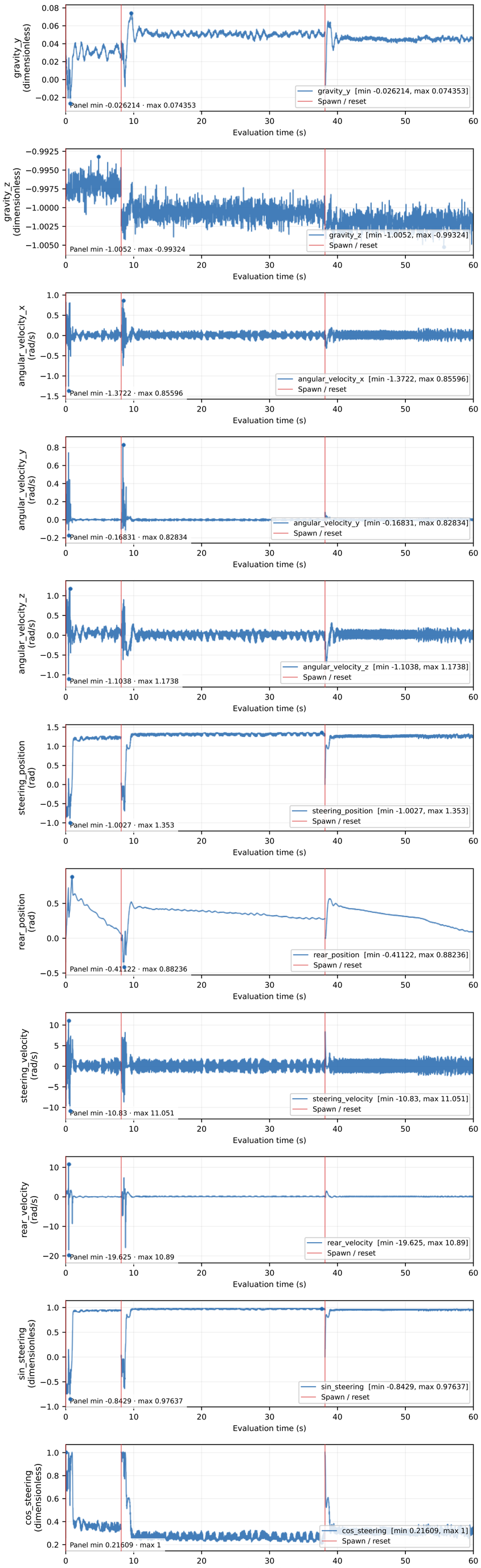
model_2500 · All 11 actual policy observations · env 56 · full 60 s

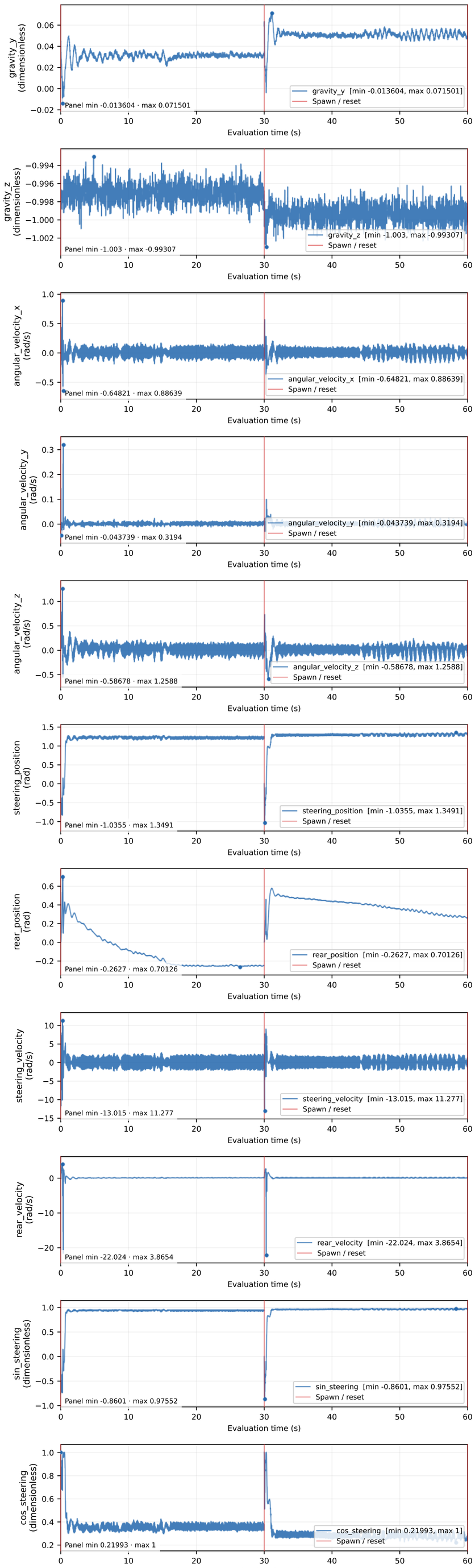
Isaac Lab · physics/PD 400 Hz · env 56
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 56
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 56
DR · 0.00–60.00 s



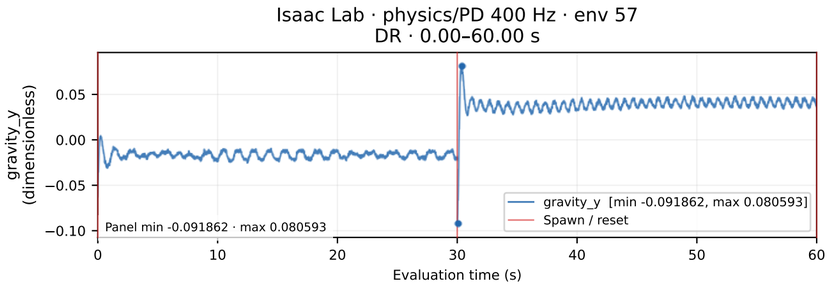
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

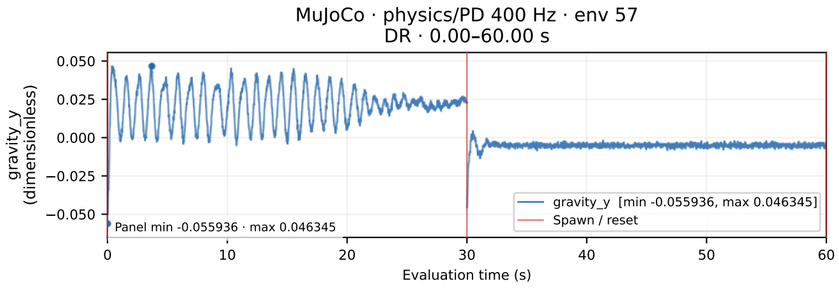
```
SAMPLED DR: env 56; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.978548, 0.933041, 0.929648, 0.983315]
  CoM XYZ mm [-3.25596, -4.31841, 2.93314]
  mu_s [0.787792, 0.628821, 0.598343, 0.785466]
  mu_d [0.617134, 0.628821, 0.598343, 0.52258]
  Kp [steer,rear] [9.19854, 0]; Kd [0.208574, 0]
  spawn roll 0.0423985 rad; rate -0.00768098 rad/s
  gyro bias XYZ rad/s [-0.000145776, -0.000196373, 5.05469e-05]
  gravity bias YZ [-0.00159637, 0.00238998]
E2 [30.00, 60.00) s:
  mass x [0.994449, 0.849617, 0.945306, 1.1639]
  CoM XYZ mm [-3.41831, -2.02272, 4.95651]
  mu_s [0.681711, 0.632409, 0.751971, 0.632409]
  mu_d [0.611756, 0.589076, 0.541224, 0.589076]
  Kp [steer,rear] [10.3618, 0]; Kd [0.195038, 0]
  spawn roll -0.0946161 rad; rate -0.231065 rad/s
  gyro bias XYZ rad/s [0.00018137, 0.000240971, 9.6531e-05]
  gravity bias YZ [0.00487678, -0.00370653]
Six significant digits shown; full precision: sampled_dr_values.json.
```

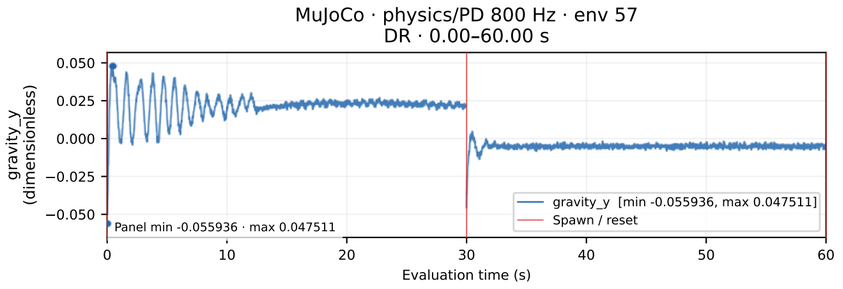
```
SAMPLED DR: env 56; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 8.16) s:
  mass x [1.17746, 1.14053, 0.82888, 0.916452]
  CoM XYZ mm [-4.13249, -1.50093, -2.20324]
  mu_s [0.538976, 0.757579, 0.559972, 0.748289]
  mu_d [0.538976, 0.70921, 0.502209, 0.689499]
  Kp [steer,rear] [9.78832, 0]; Kd [0.18094, 0]
  spawn roll -0.0446931 rad; rate -0.236748 rad/s
  gyro bias XYZ rad/s [-1.33536e-05, -0.000180171, 9.54982e-05]
  gravity bias YZ [-0.00148982, 0.00237804]
E2 [8.16, 38.16) s:
  mass x [0.856354, 0.811058, 1.04212, 1.0589]
  CoM XYZ mm [0.588333, -3.77611, -0.66241]
  mu_s [0.640016, 0.597748, 0.631146, 0.559972]
  mu_d [0.513141, 0.597748, 0.631146, 0.502209]
  Kp [steer,rear] [10.7806, 0]; Kd [0.183061, 0]
  spawn roll -0.0436161 rad; rate 0.0759602 rad/s
  gyro bias XYZ rad/s [-4.65311e-05, -0.000153393, 7.44427e-05]
  gravity bias YZ [-0.00122513, -0.00191494]
E3 [38.16, 60.00) s:
  mass x [1.13284, 0.807767, 1.00117, 1.19138]
  CoM XYZ mm [-0.178846, -2.61705, 3.94731]
  mu_s [0.700521, 0.666375, 0.71155, 0.666375]
  mu_d [0.641329, 0.519145, 0.71155, 0.519145]
  Kp [steer,rear] [9.39402, 0]; Kd [0.182916, 0]
  spawn roll 0.0244649 rad; rate 0.0222008 rad/s
  gyro bias XYZ rad/s [-0.000224125, 0.000416155, 0.000182462]
  gravity bias YZ [-0.00161299, -0.00305434]
Six significant digits shown; full precision: sampled_dr_values.json.
```

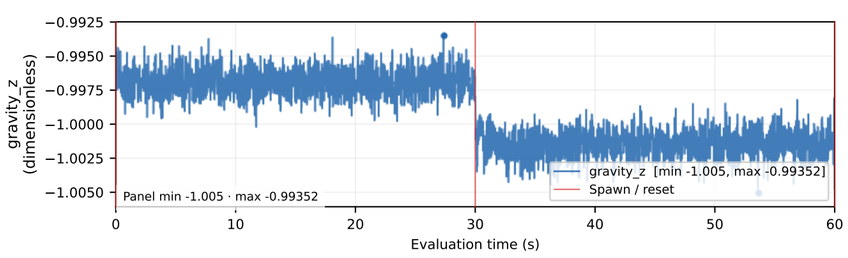
```
SAMPLED DR: env 56; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17746, 1.14053, 0.82888, 0.916452]
  CoM XYZ mm [-4.13249, -1.50093, -2.20324]
  mu_s [0.538976, 0.757579, 0.559972, 0.748289]
  mu_d [0.538976, 0.70921, 0.502209, 0.689499]
  Kp [steer,rear] [9.78832, 0]; Kd [0.18094, 0]
  spawn roll -0.0446931 rad; rate -0.236748 rad/s
  gyro bias XYZ rad/s [-1.33536e-05, -0.000180171, 9.54982e-05]
  gravity bias YZ [-0.00148982, 0.00237804]
E2 [30.00, 60.00) s:
  mass x [1.08147, 1.04442, 1.08994, 1.11074]
  CoM XYZ mm [-3.25819, -4.05595, 1.49957]
  mu_s [0.666375, 0.631146, 0.633024, 0.727426]
  mu_d [0.519145, 0.631146, 0.568172, 0.606358]
  Kp [steer,rear] [10.354, 0]; Kd [0.193614, 0]
  spawn roll -0.064361 rad; rate -0.0590807 rad/s
  gyro bias XYZ rad/s [0.000253148, 0.000173571, -0.000125265]
  gravity bias YZ [-0.00262255, -0.00076504]
Six significant digits shown; full precision: sampled_dr_values.json.
```

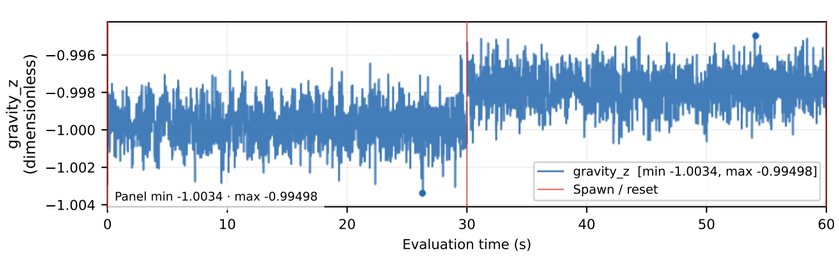
model_2500 · All 11 actual policy observations · env 57 · full 60 s

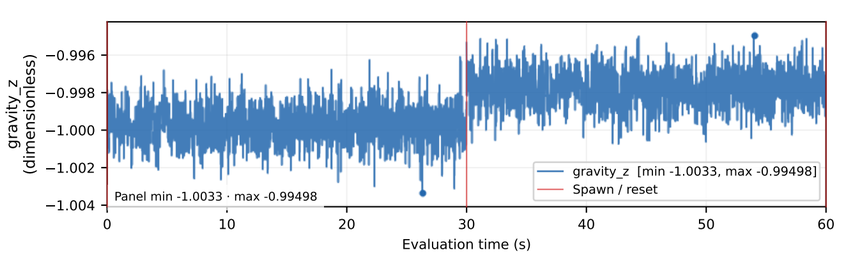


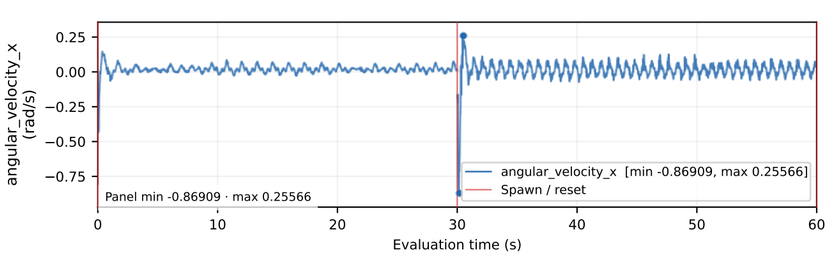


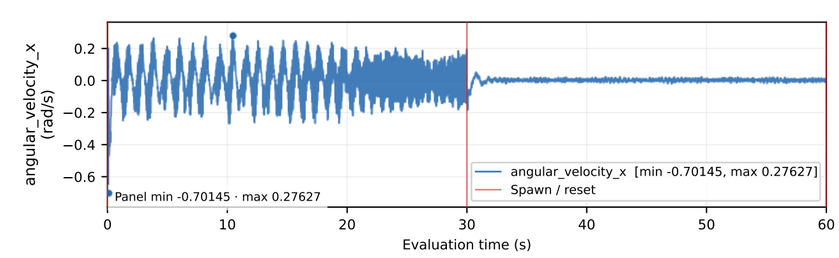


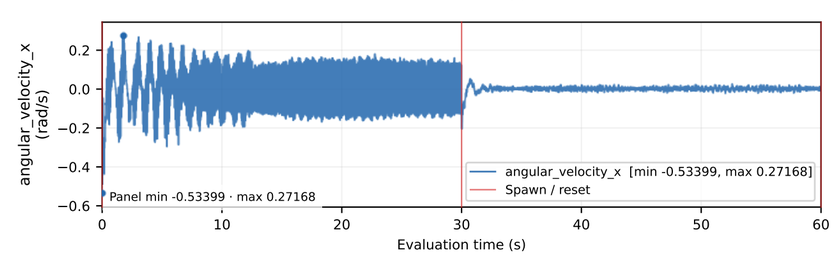


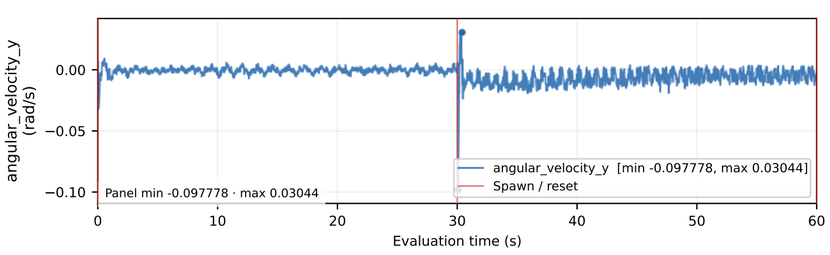


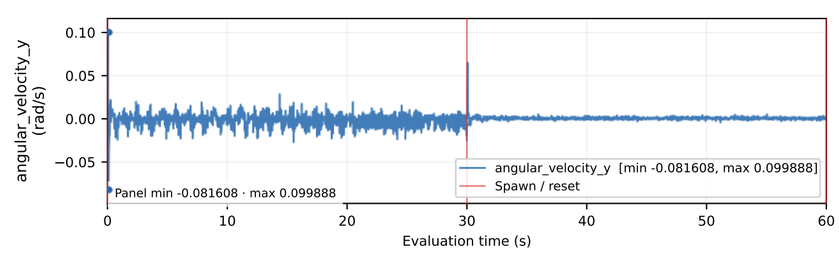


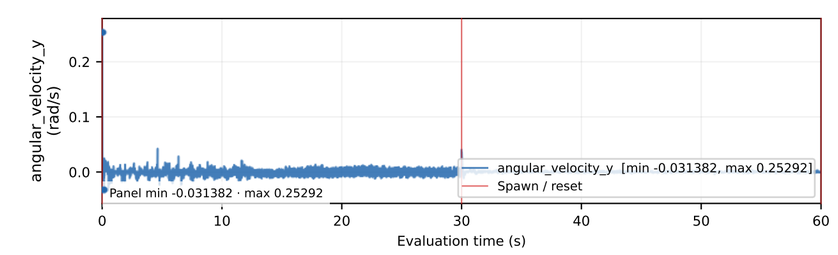


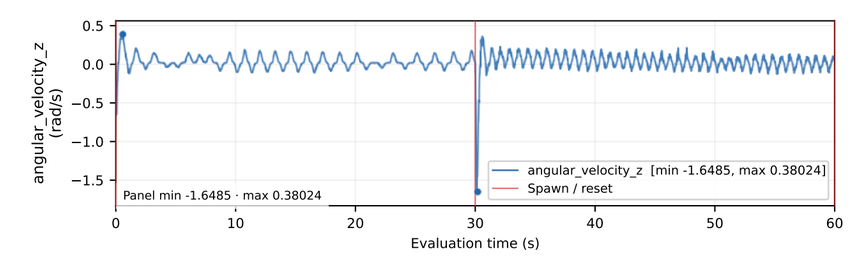


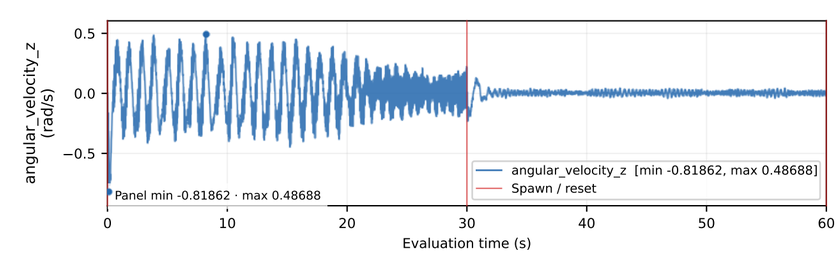


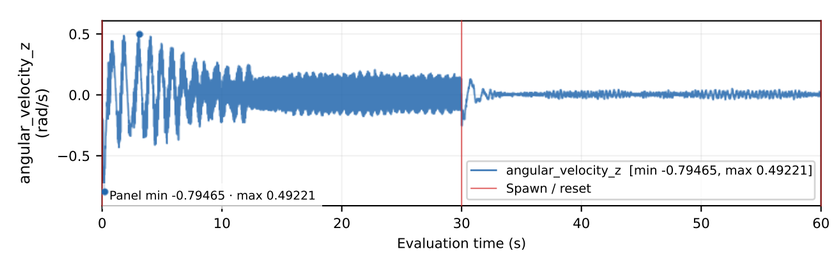


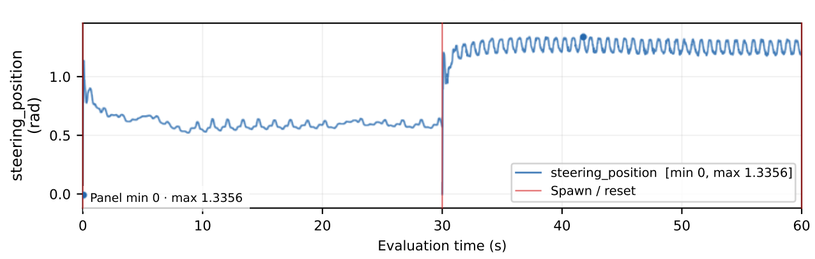


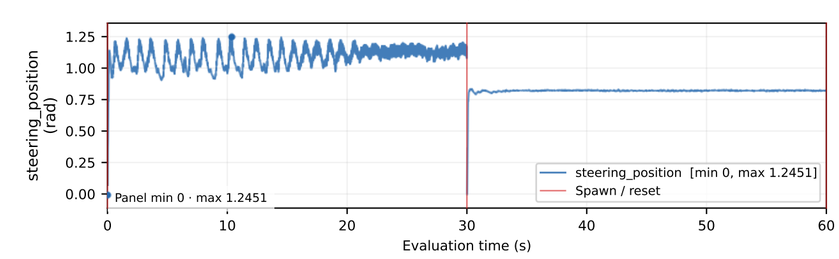


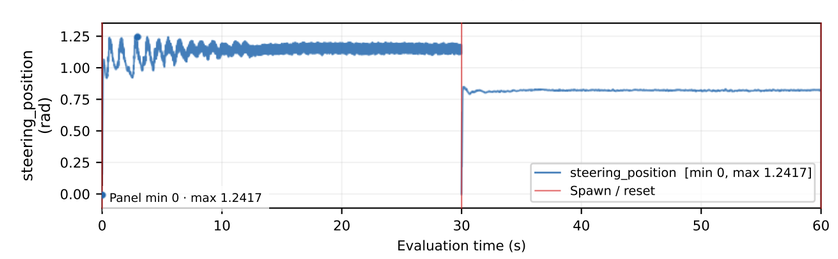


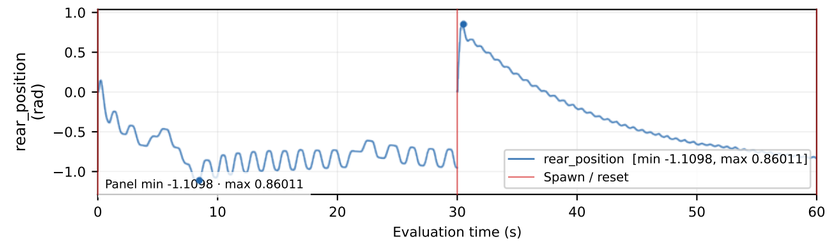


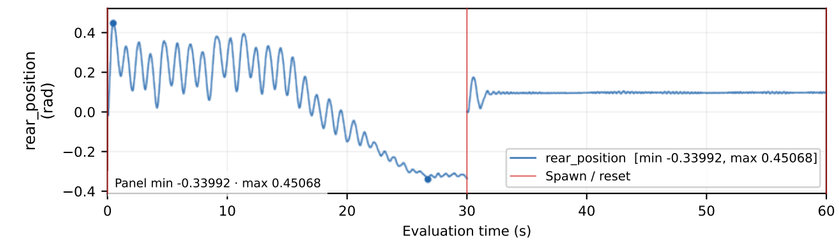


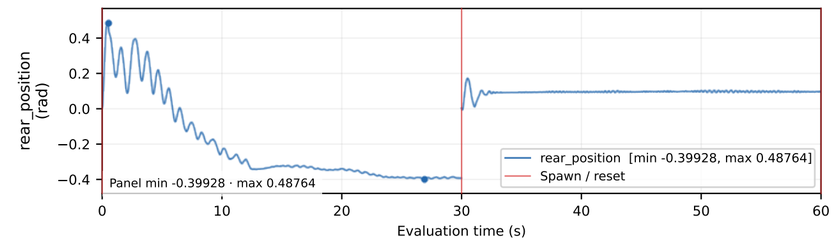


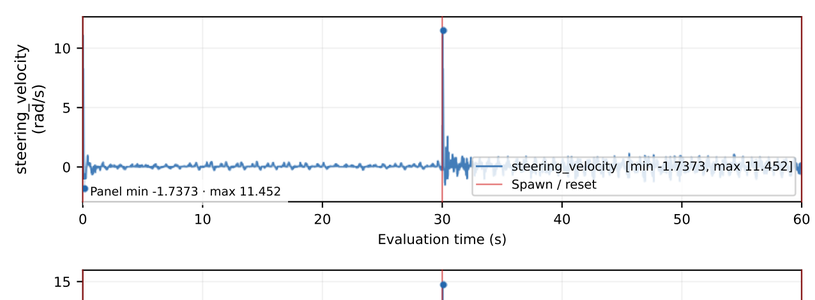


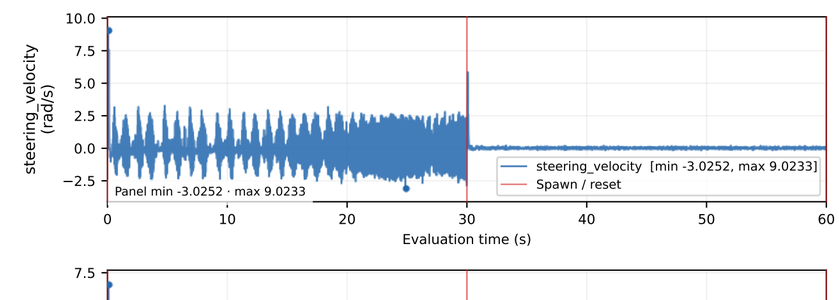


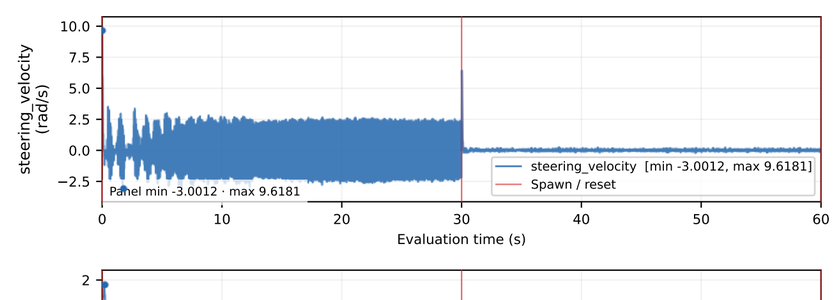


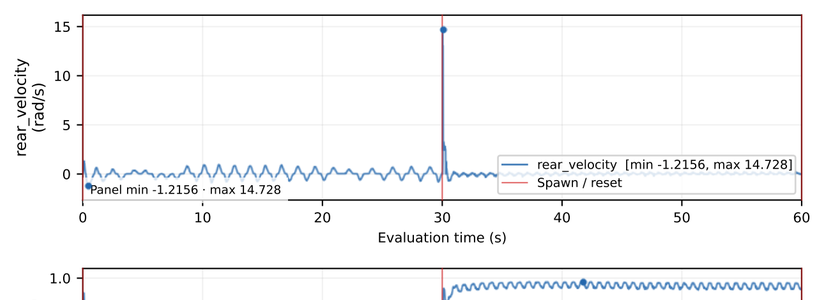


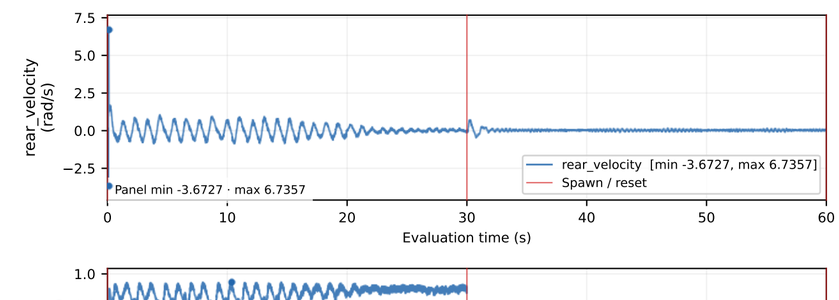


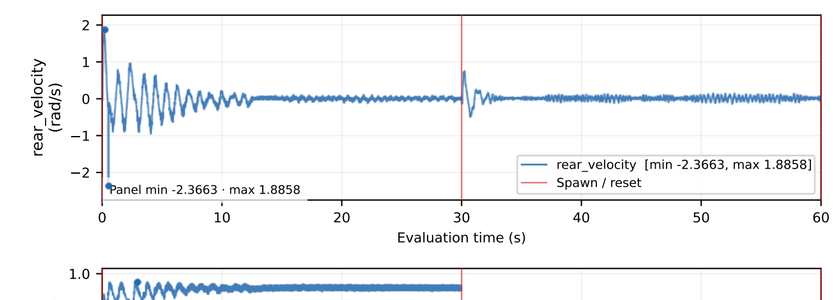


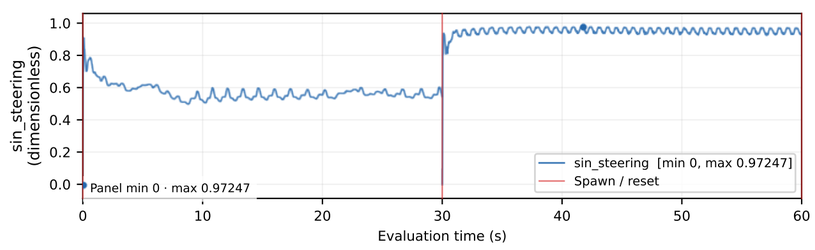


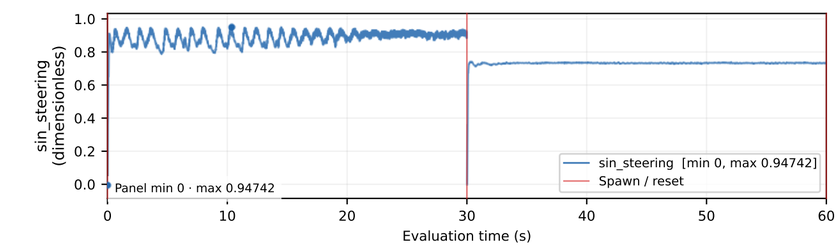


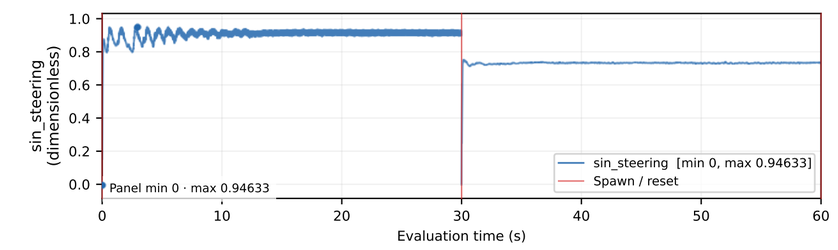


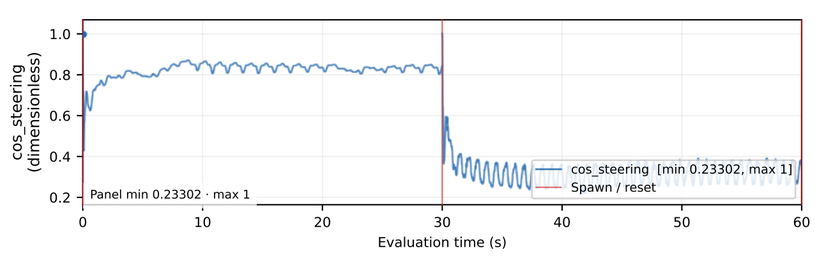


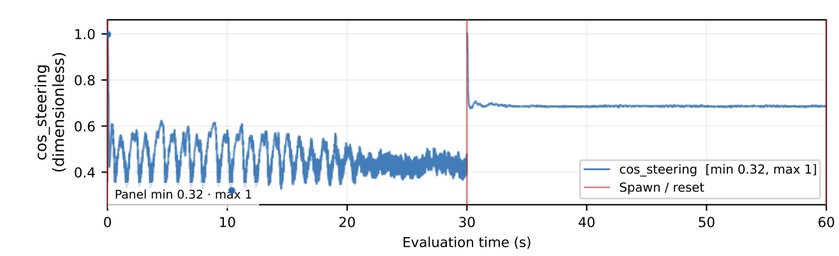


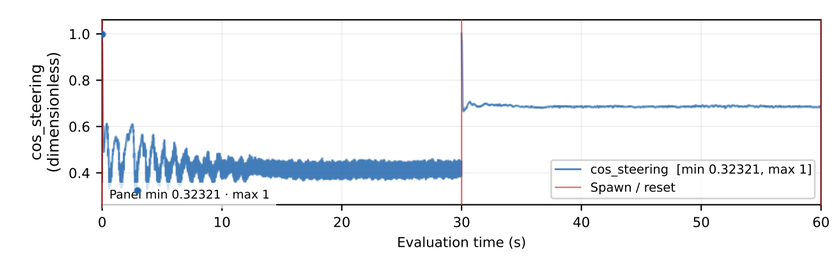












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 57; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03432, 0.881207, 1.02932, 1.18585]
  CoM XYZ mm [-0.983864, 4.94792, -4.15209]
  mu_s [0.681711, 0.584424, 0.584424, 0.764681]
  mu_d [0.611756, 0.584424, 0.584424, 0.764681]
  Kp [steer,rear] [9.40944, 0]; Kd [0.182484, 0]
  spawn roll 0.0620867 rad; rate -0.00318035 rad/s
  gyro bias XYZ rad/s [0.000308029, 0.000200594, 0.000217675]
  gravity bias YZ [-0.00138079, 0.0029808]
E2 [30.00, 60.00) s:
  mass x [0.854298, 1.007, 1.06667, 0.956546]
  CoM XYZ mm [-3.08383, -2.12793, -2.92704]
  mu_s [0.794338, 0.531594, 0.787792, 0.713129]
  mu_d [0.526209, 0.531594, 0.617134, 0.713129]
  Kp [steer,rear] [9.88376, 0]; Kd [0.20941, 0]
  spawn roll 0.091352 rad; rate -0.174524 rad/s
  gyro bias XYZ rad/s [-3.3195e-05, -5.78408e-05, -5.78026e-05]
  gravity bias YZ [-0.00101303, -0.00231498]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 57; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.00241, 1.02604, 1.00477, 1.18887]
  CoM XYZ mm [1.14903, 0.682835, -2.13213]
  mu_s [0.700521, 0.640867, 0.791209, 0.546287]
  mu_d [0.641329, 0.556841, 0.767936, 0.546287]
  Kp [steer,rear] [10.2201, 0]; Kd [0.189835, 0]
  spawn roll 0.0549298 rad; rate -0.229716 rad/s
  gyro bias XYZ rad/s [5.2601e-05, 6.57036e-05, 0.000187049]
  gravity bias YZ [-0.00175523, -9.17922e-05]
E2 [30.00, 60.00) s:
  mass x [1.08541, 0.924102, 1.11539, 1.05289]
  CoM XYZ mm [-3.29912, 4.5428, 1.20135]
  mu_s [0.559972, 0.71155, 0.640016, 0.631146]
  mu_d [0.502209, 0.71155, 0.513141, 0.631146]
  Kp [steer,rear] [10.8444, 0]; Kd [0.210072, 0]
  spawn roll 0.0469243 rad; rate -0.131273 rad/s
  gyro bias XYZ rad/s [-0.000102709, 0.00027766, -1.37505e-05]
  gravity bias YZ [0.00161779, 0.00214689]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 57; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.00241, 1.02604, 1.00477, 1.18887]
  CoM XYZ mm [1.14903, 0.682835, -2.13213]
  mu_s [0.700521, 0.640867, 0.791209, 0.546287]
  mu_d [0.641329, 0.556841, 0.767936, 0.546287]
  Kp [steer,rear] [10.2201, 0]; Kd [0.189835, 0]
  spawn roll 0.0549298 rad; rate -0.229716 rad/s
  gyro bias XYZ rad/s [5.2601e-05, 6.57036e-05, 0.000187049]
  gravity bias YZ [-0.00175523, -9.17922e-05]
E2 [30.00, 60.00) s:
  mass x [1.08541, 0.924102, 1.11539, 1.05289]
  CoM XYZ mm [-3.29912, 4.5428, 1.20135]
  mu_s [0.559972, 0.71155, 0.640016, 0.631146]
  mu_d [0.502209, 0.71155, 0.513141, 0.631146]
  Kp [steer,rear] [10.8444, 0]; Kd [0.210072, 0]
  spawn roll 0.0469243 rad; rate -0.131273 rad/s
  gyro bias XYZ rad/s [-0.000102709, 0.00027766, -1.37505e-05]
  gravity bias YZ [0.00161779, 0.00214689]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 58 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 58
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 58
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 58
DR · 0.00–60.00 s

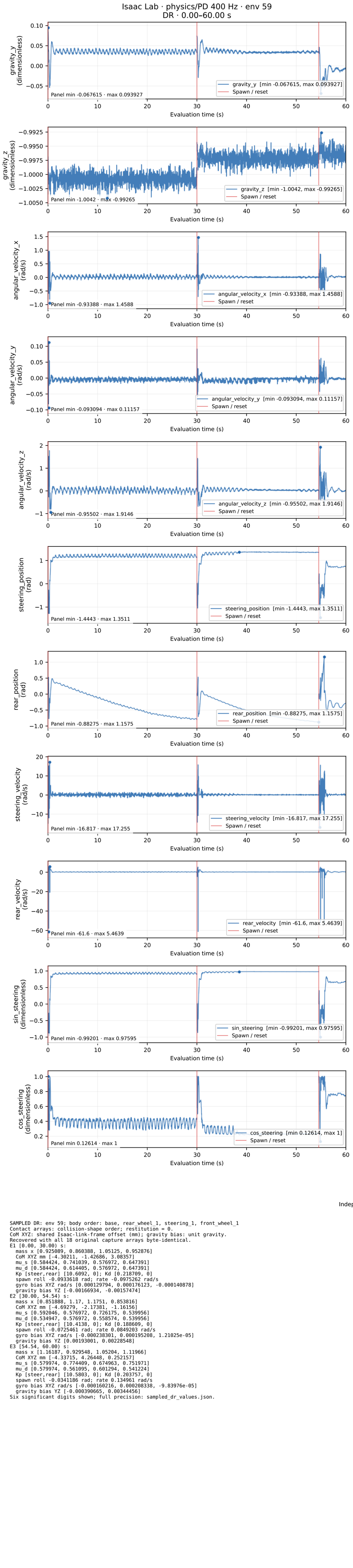
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

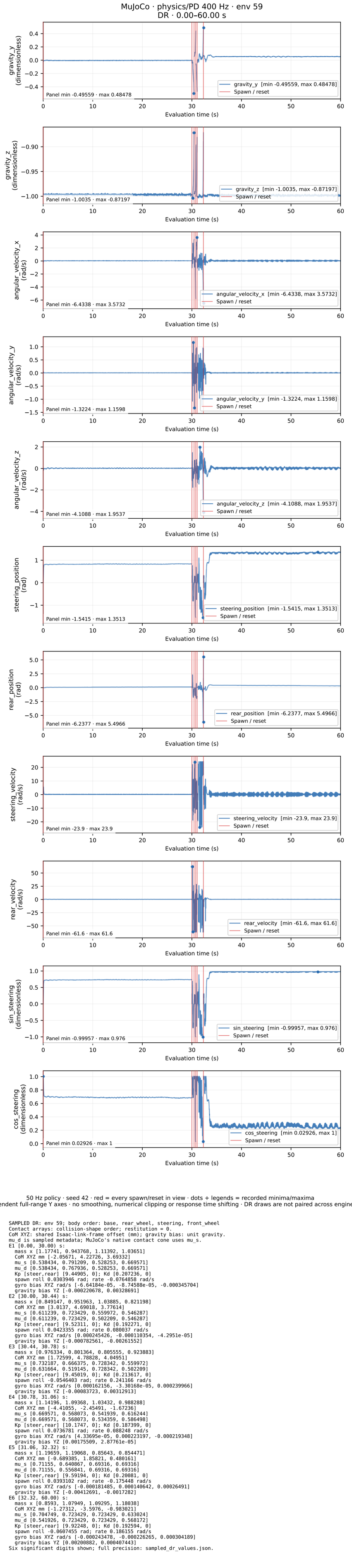
```
SAMPLED DR: env 58; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 2.48) s:
  mass x [1.18769, 1.06116, 1.01873, 0.96856]
  CoM XYZ mm [4.84164, 4.93649, -1.37092]
  mu_s [0.531594, 0.547743, 0.703711, 0.628821]
  mu_d [0.531594, 0.547743, 0.703711, 0.628821]
  Kp [steer,rear] [10.3875, 0]; Kd [0.217806, 0]
  spawn roll -0.0226035 rad; rate 0.0441529 rad/s
  gyro bias XYZ rad/s [0.000287679, 0.000107617, -1.96523e-06]
  gravity bias YZ [-0.00102433, -0.00207661]
E2 [2.48, 32.48) s:
  mass x [1.18104, 0.895951, 0.89566, 1.1683]
  CoM XYZ mm [1.12556, -1.30424, 3.61978]
  mu_s [0.673378, 0.681711, 0.602694, 0.647391]
  mu_d [0.673378, 0.611756, 0.602694, 0.647391]
  Kp [steer,rear] [9.22221, 0]; Kd [0.186914, 0]
  spawn roll -0.0732774 rad; rate -0.210146 rad/s
  gyro bias XYZ rad/s [0.000170324, -0.000262011, 0.000160037]
  gravity bias YZ [-0.00118923, 0.000579164]
E3 [32.48, 60.00) s:
  mass x [1.05184, 0.975285, 0.958412, 0.853304]
  CoM XYZ mm [-4.75869, -2.26922, -2.61918]
  mu_s [0.659445, 0.751971, 0.681711, 0.584424]
  mu_d [0.547619, 0.541224, 0.611756, 0.584424]
  Kp [steer,rear] [9.22393, 0]; Kd [0.1872, 0]
  spawn roll 0.0823566 rad; rate 0.196909 rad/s
  gyro bias XYZ rad/s [9.81637e-05, -0.000120994, 3.03913e-05]
  gravity bias YZ [3.49841e-06, 0.00121471]
Six significant digits shown; full precision: sampled_dr_values.json.
```

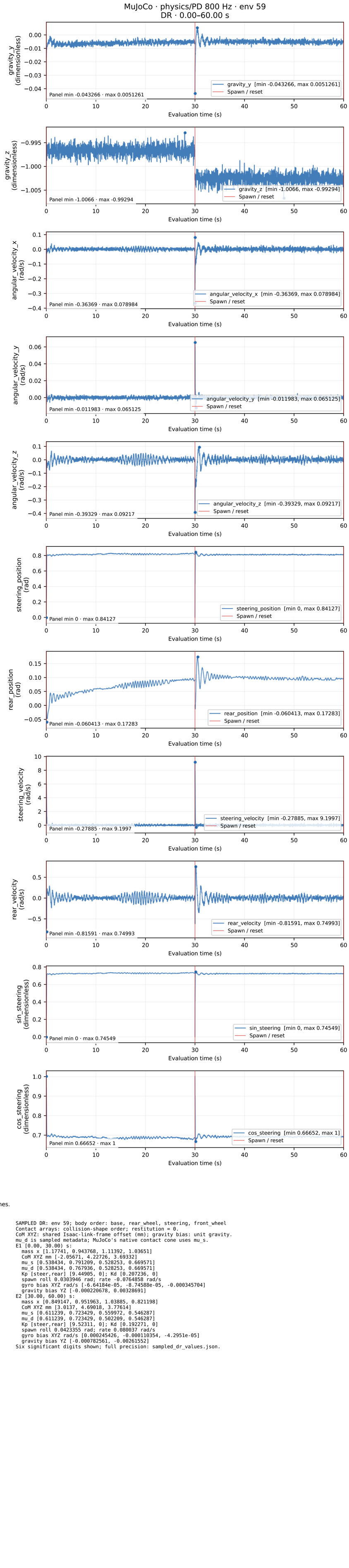
```
SAMPLED DR: env 58; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [1.13399, 1.03862, 0.915545, 0.817181]
  CoM XYZ mm [4.73654, 0.964717, 2.90263]
  mu_s [0.541939, 0.541939, 0.732187, 0.749678]
  mu_d [0.534359, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [9.37998, 0]; Kd [0.19139, 0]
  spawn roll 0.0542045 rad; rate -0.00291911 rad/s
  gyro bias XYZ rad/s [0.000350847, -0.000267596, 6.51149e-05]
  gravity bias YZ [-0.00137824, -3.96436e-05]
E2 [0.44, 30.44) s:
  mass x [1.18822, 0.970738, 0.849511, 1.0935]
  CoM XYZ mm [3.21299, 0.60756, -0.358027]
  mu_s [0.538976, 0.633024, 0.559972, 0.700521]
  mu_d [0.538976, 0.568172, 0.502209, 0.641329]
  Kp [steer,rear] [10.7819, 0]; Kd [0.185642, 0]
  spawn roll 0.0740205 rad; rate 0.0585414 rad/s
  gyro bias XYZ rad/s [0.000159138, -0.000206941, -0.00018457]
  gravity bias YZ [0.00128601, -0.00104469]
E3 [30.44, 60.00) s:
  mass x [0.937507, 0.860351, 1.10632, 1.16687]
  CoM XYZ mm [2.00202, -2.7896, 0.884465]
  mu_s [0.723429, 0.727426, 0.637675, 0.631146]
  mu_d [0.723429, 0.606358, 0.637675, 0.631146]
  Kp [steer,rear] [9.60298, 0]; Kd [0.20182, 0]
  spawn roll 0.0333651 rad; rate 0.233172 rad/s
  gyro bias XYZ rad/s [-5.82319e-05, -0.000232986, 0.00010145]
  gravity bias YZ [-0.00382904, 0.000744485]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 58; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.13399, 1.03862, 0.915545, 0.817181]
  CoM XYZ mm [4.73654, 0.964717, 2.90263]
  mu_s [0.541939, 0.541939, 0.732187, 0.749678]
  mu_d [0.534359, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [9.37998, 0]; Kd [0.19139, 0]
  spawn roll 0.0542045 rad; rate -0.00291911 rad/s
  gyro bias XYZ rad/s [0.000350847, -0.000267596, 6.51149e-05]
  gravity bias YZ [-0.00137824, -3.96436e-05]
E2 [30.00, 60.00) s:
  mass x [0.95846, 1.07212, 0.934933, 1.07707]
  CoM XYZ mm [-2.64052, -4.04246, 3.64696]
  mu_s [0.71008, 0.637675, 0.559972, 0.666375]
  mu_d [0.59371, 0.637675, 0.582209, 0.519145]
  Kp [steer,rear] [9.89135, 0]; Kd [0.209232, 0]
  spawn roll 0.00937321 rad; rate -0.152752 rad/s
  gyro bias XYZ rad/s [-5.94112e-06, -0.000154327, 1.45148e-05]
  gravity bias YZ [-0.0028596, -0.00716214]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 59 · full 60 s







50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

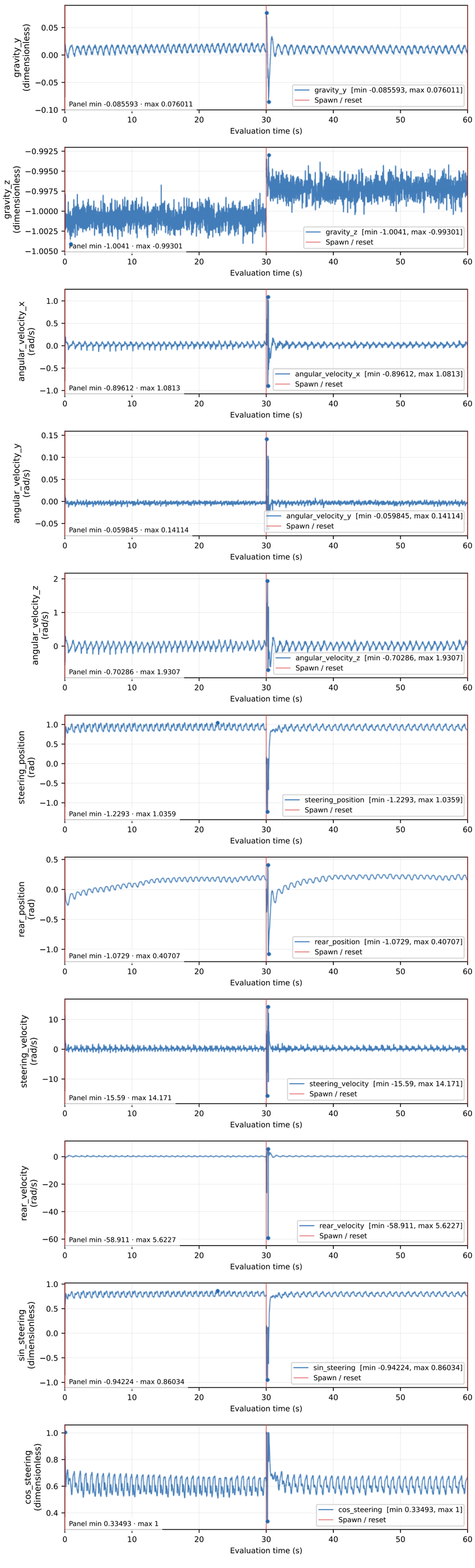
```
SAMPLED DR: env 59; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.925089, 0.860388, 1.05125, 0.952876]
  CoM XYZ mm [-4.30211, -1.42686, 3.08357]
  mu_s [0.584424, 0.741039, 0.576972, 0.647391]
  mu_d [0.584424, 0.614405, 0.576972, 0.647391]
  Kp [steer,rear] [10.6092, 0]; Kd [0.218709, 0]
  spawn roll -0.0933618 rad; rate -0.0975262 rad/s
  gyro bias XYZ rad/s [0.000129794, 0.000176123, -0.000140878]
  gravity bias YZ [-0.00166934, -0.00157474]
E2 [30.00, 54.54) s:
  mass x [0.851888, 1.17, 1.1751, 0.853816]
  CoM XYZ mm [-4.69279, -2.17381, -1.16156]
  mu_s [0.592046, 0.576972, 0.726175, 0.539956]
  mu_d [0.534947, 0.576972, 0.558574, 0.539956]
  Kp [steer,rear] [10.4138, 0]; Kd [0.188609, 0]
  spawn roll -0.0725461 rad; rate 0.0849203 rad/s
  gyro bias XYZ rad/s [-0.000238301, 0.000195208, 1.21025e-05]
  gravity bias YZ [0.00193001, 0.00228548]
E3 [54.54, 60.00) s:
  mass x [1.16187, 0.929548, 1.05204, 1.11966]
  CoM XYZ mm [-4.33715, 4.26448, 0.252157]
  mu_s [0.579974, 0.774409, 0.674963, 0.751971]
  mu_d [0.579974, 0.561095, 0.601294, 0.541224]
  Kp [steer,rear] [10.5803, 0]; Kd [0.203757, 0]
  spawn roll -0.0341186 rad; rate 0.134961 rad/s
  gyro bias XYZ rad/s [-0.000160216, 0.000208338, -9.83976e-05]
  gravity bias YZ [-0.000390665, 0.00344456]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 59; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17741, 0.943768, 1.11392, 1.03651]
  CoM XYZ mm [-2.05671, 4.22726, 3.69332]
  mu_s [0.538434, 0.791209, 0.528253, 0.669571]
  mu_d [0.538434, 0.767936, 0.528253, 0.669571]
  Kp [steer,rear] [9.44905, 0]; Kd [0.207236, 0]
  spawn roll 0.0303946 rad; rate -0.0764858 rad/s
  gyro bias XYZ rad/s [-6.64184e-05, -8.74588e-05, -0.000345704]
  gravity bias YZ [-0.000220678, 0.00328691]
E2 [30.00, 30.44) s:
  mass x [0.849147, 0.951963, 1.03885, 0.821198]
  CoM XYZ mm [3.0137, 4.69018, 3.77614]
  mu_s [0.611239, 0.723429, 0.559972, 0.546287]
  mu_d [0.611239, 0.723429, 0.502209, 0.546287]
  Kp [steer,rear] [9.52311, 0]; Kd [0.192271, 0]
  spawn roll 0.0423355 rad; rate 0.080037 rad/s
  gyro bias XYZ rad/s [0.000245426, -0.000110354, -4.2951e-05]
  gravity bias YZ [-0.000782561, -0.00261552]
E3 [30.44, 30.78) s:
  mass x [0.976334, 0.801364, 0.805555, 0.923883]
  CoM XYZ mm [1.72599, 4.78828, 4.04951]
  mu_s [0.732187, 0.666375, 0.728342, 0.559972]
  mu_d [0.631664, 0.519145, 0.728342, 0.502209]
  Kp [steer,rear] [9.45019, 0]; Kd [0.213617, 0]
  spawn roll -0.0546403 rad; rate 0.241166 rad/s
  gyro bias XYZ rad/s [0.000162156, -3.30168e-05, 0.000239966]
  gravity bias YZ [-0.00083723, 0.00312913]
E4 [30.78, 31.06) s:
  mass x [1.14196, 1.09368, 1.03432, 0.988288]
  CoM XYZ mm [-4.41055, -2.45491, -1.67236]
  mu_s [0.669571, 0.568073, 0.541939, 0.616244]
  mu_d [0.669571, 0.568073, 0.534359, 0.586498]
  Kp [steer,rear] [10.1747, 0]; Kd [0.187399, 0]
  spawn roll 0.0736781 rad; rate 0.088248 rad/s
  gyro bias XYZ rad/s [4.33695e-05, 0.000223197, -0.000219348]
  gravity bias YZ [0.00175509, 2.87761e-05]
E5 [31.06, 32.32) s:
  mass x [1.19659, 1.19068, 0.85643, 0.854471]
  CoM XYZ mm [-0.689385, 1.85821, 0.480161]
  mu_s [0.71155, 0.640867, 0.69316, 0.69316]
  mu_d [0.71155, 0.556841, 0.69316, 0.69316]
  Kp [steer,rear] [9.59194, 0]; Kd [0.20081, 0]
  spawn roll 0.0393102 rad; rate -0.175448 rad/s
  gyro bias XYZ rad/s [-0.000181485, 0.000140642, 0.00026491]
  gravity bias YZ [-0.00412691, -0.0017282]
E6 [32.32, 60.00) s:
  mass x [0.8593, 1.07949, 1.09295, 1.18038]
  CoM XYZ mm [-1.27312, -3.5976, -0.983021]
  mu_s [0.704749, 0.723429, 0.723429, 0.633024]
  mu_d [0.541926, 0.723429, 0.723429, 0.568172]
  Kp [steer,rear] [9.92248, 0]; Kd [0.192594, 0]
  spawn roll -0.0607455 rad; rate 0.186155 rad/s
  gyro bias XYZ rad/s [-0.000243478, -0.000226265, 0.000304189]
  gravity bias YZ [0.00200882, 0.000407443]
Six significant digits shown; full precision: sampled_dr_values.json.
```

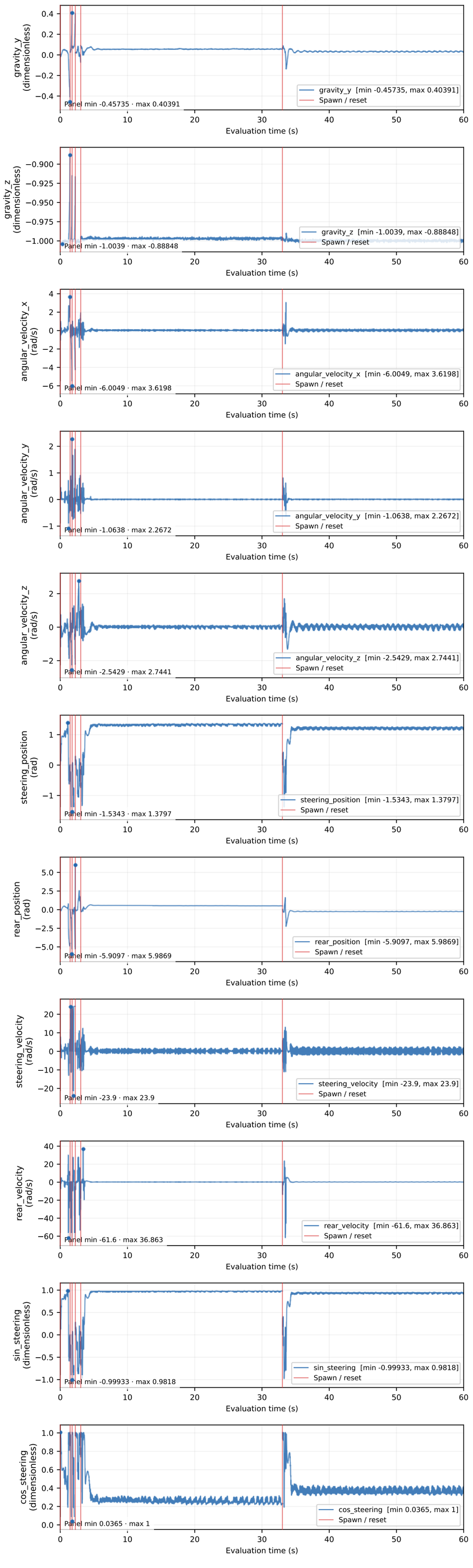
```
SAMPLED DR: env 59; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17741, 0.943768, 1.11392, 1.03651]
  CoM XYZ mm [-2.05671, 4.22726, 3.69332]
  mu_s [0.538434, 0.791209, 0.528253, 0.669571]
  mu_d [0.538434, 0.767936, 0.528253, 0.669571]
  Kp [steer,rear] [9.44905, 0]; Kd [0.207236, 0]
  spawn roll 0.0303946 rad; rate -0.0764858 rad/s
  gyro bias XYZ rad/s [-6.64184e-05, -8.74588e-05, -0.000345704]
  gravity bias YZ [-0.000220678, 0.00328691]
E2 [30.00, 60.00) s:
  mass x [0.849147, 0.951963, 1.03885, 0.821198]
  CoM XYZ mm [3.0137, 4.69018, 3.77614]
  mu_s [0.611239, 0.723429, 0.559972, 0.546287]
  mu_d [0.611239, 0.723429, 0.502209, 0.546287]
  Kp [steer,rear] [9.52311, 0]; Kd [0.192271, 0]
  spawn roll 0.0423355 rad; rate 0.080037 rad/s
  gyro bias XYZ rad/s [0.000245426, -0.000110354, -4.2951e-05]
  gravity bias YZ [-0.000782561, -0.00261552]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 60 · full 60 s

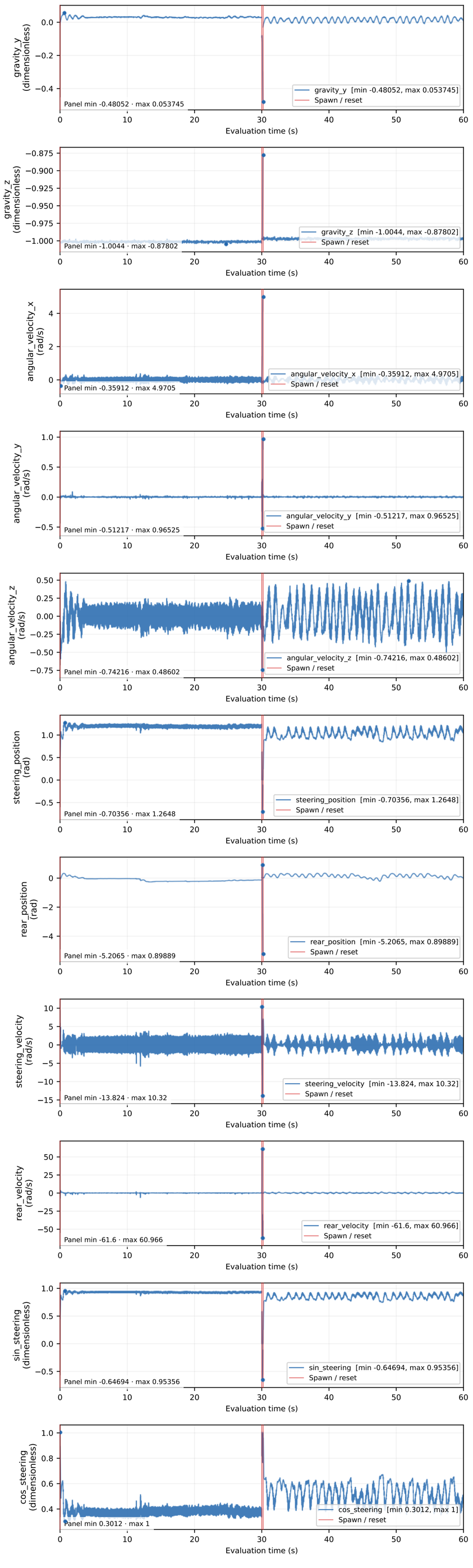
Isaac Lab · physics/PD 400 Hz · env 60
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 60
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 60
DR · 0.00–60.00 s



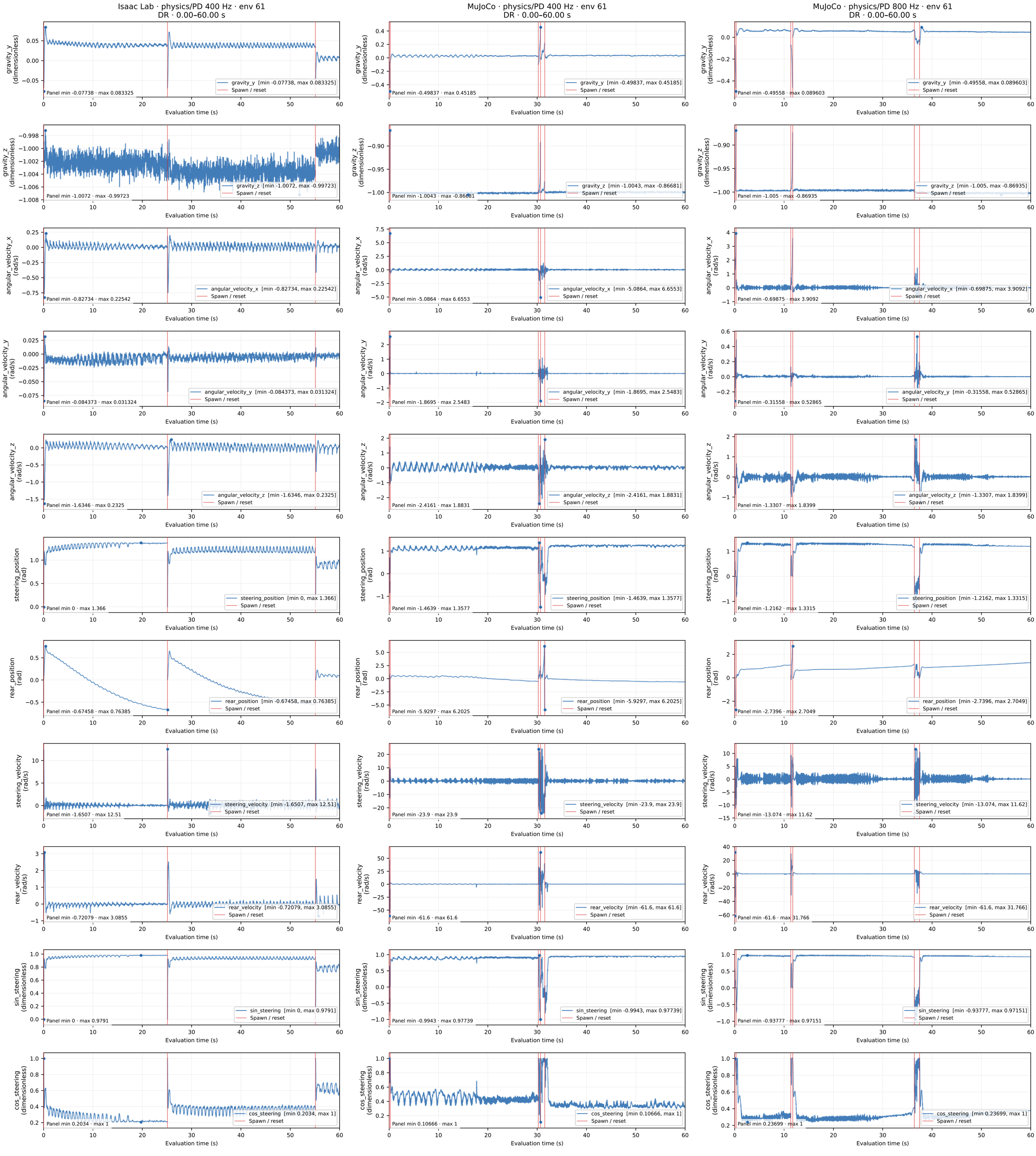
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 60; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.900482, 1.10467, 0.935847, 1.18135]
  CoM XYZ mm [-3.37732, 2.48318, 4.58559]
  mu_s [0.547743, 0.703711, 0.539956, 0.693205]
  mu_d [0.547743, 0.703711, 0.539956, 0.693205]
  Kp [steer,rear] [10.9047, 0]; Kd [0.182122, 0]
  spawn roll -0.00452483 rad; rate 0.138263 rad/s
  gyro bias XYZ rad/s [0.000207397, 1.01754e-05, -0.000280071]
  gravity bias YZ [0.00226908, -0.000901314]
E2 [30.00, 60.00) s:
  mass x [0.800306, 1.12363, 0.945733, 0.825305]
  CoM XYZ mm [4.75881, 2.31119, 3.9894]
  mu_s [0.533455, 0.539956, 0.602694, 0.531594]
  mu_d [0.533455, 0.539956, 0.602694, 0.531594]
  Kp [steer,rear] [9.1609, 0]; Kd [0.207236, 0]
  spawn roll -0.076851 rad; rate 0.24644 rad/s
  gyro bias XYZ rad/s [0.000419029, 0.000123381, -0.000165383]
  gravity bias YZ [-0.000913082, 0.00265855]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 60; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.46) s:
  mass x [0.863996, 1.03438, 1.12953, 0.918904]
  CoM XYZ mm [1.849, -0.361158, 3.91357]
  mu_s [0.559972, 0.538976, 0.568073, 0.669571]
  mu_d [0.502209, 0.538976, 0.568073, 0.669571]
  Kp [steer,rear] [9.48504, 0]; Kd [0.193822, 0]
  spawn roll 0.0180142 rad; rate 0.112822 rad/s
  gyro bias XYZ rad/s [-0.000120723, -7.37988e-05, 0.000199168]
  gravity bias YZ [-0.000972708, -0.00177809]
E2 [1.46, 1.76) s:
  mass x [0.917893, 1.07381, 0.843499, 0.825065]
  CoM XYZ mm [-2.4307, 0.779571, 4.62948]
  mu_s [0.538976, 0.727426, 0.640016, 0.528253]
  mu_d [0.538976, 0.606358, 0.513141, 0.528253]
  Kp [steer,rear] [10.581, 0]; Kd [0.194328, 0]
  spawn roll -0.0303228 rad; rate -0.212432 rad/s
  gyro bias XYZ rad/s [-0.000131211, -0.000202156, -0.000439974]
  gravity bias YZ [-0.00121586, -0.000284463]
E3 [1.76, 2.24) s:
  mass x [1.1042, 1.1535, 0.911097, 0.820985]
  CoM XYZ mm [-1.70354, -1.84323, -2.08665]
  mu_s [0.669571, 0.631146, 0.71008, 0.633024]
  mu_d [0.669571, 0.631146, 0.59371, 0.568172]
  Kp [steer,rear] [10.4149, 0]; Kd [0.204249, 0]
  spawn roll -0.0748894 rad; rate -0.0321481 rad/s
  gyro bias XYZ rad/s [7.76e-05, 0.000126532, 0.000184117]
  gravity bias YZ [-0.00145927, -0.00195968]
E4 [2.24, 3.04) s:
  mass x [0.874368, 1.03209, 0.827202, 1.11763]
  CoM XYZ mm [1.8891, 2.49642, -1.97656]
  mu_s [0.616244, 0.640016, 0.538976, 0.633024]
  mu_d [0.586498, 0.513141, 0.538976, 0.568172]
  Kp [steer,rear] [9.23719, 0]; Kd [0.219542, 0]
  spawn roll -0.0172836 rad; rate 0.236369 rad/s
  gyro bias XYZ rad/s [2.82859e-06, -4.40666e-05, -0.000137117]
  gravity bias YZ [0.00154996, -0.00150121]
E5 [3.04, 33.04) s:
  mass x [1.17026, 0.851044, 1.05302, 1.05592]
  CoM XYZ mm [-1.43378, -4.5833, -2.38461]
  mu_s [0.666375, 0.728342, 0.723429, 0.538976]
  mu_d [0.519145, 0.728342, 0.723429, 0.538976]
  Kp [steer,rear] [10.7656, 0]; Kd [0.18096, 0]
  spawn roll -0.092104 rad; rate 0.219224 rad/s
  gyro bias XYZ rad/s [-0.000116992, -0.000102171, -0.000155466]
  gravity bias YZ [-0.0035581, 0.00140918]
E6 [33.04, 60.00) s:
  mass x [0.989861, 1.10855, 1.03997, 1.10907]
  CoM XYZ mm [2.49538, -0.314775, 1.84938]
  mu_s [0.700521, 0.71008, 0.700521, 0.727426]
  mu_d [0.641329, 0.59371, 0.641329, 0.606358]
  Kp [steer,rear] [9.94775, 0]; Kd [0.187628, 0]
  spawn roll -0.0658516 rad; rate 0.232003 rad/s
  gyro bias XYZ rad/s [-0.0003429, -3.6408e-05, -9.68509e-05]
  gravity bias YZ [0.00214218, -0.000116128]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 60; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.863996, 1.03438, 1.12953, 0.918904]
  CoM XYZ mm [1.849, -0.361158, 3.91357]
  mu_s [0.559972, 0.538976, 0.568073, 0.669571]
  mu_d [0.502209, 0.538976, 0.568073, 0.669571]
  Kp [steer,rear] [9.48504, 0]; Kd [0.193822, 0]
  spawn roll 0.0180142 rad; rate 0.112822 rad/s
  gyro bias XYZ rad/s [-0.000120723, -7.37988e-05, 0.000199168]
  gravity bias YZ [-0.000972708, -0.00177809]
E2 [30.00, 30.20) s:
  mass x [0.904864, 1.15044, 1.1004, 1.0186]
  CoM XYZ mm [0.1525, -3.43279, -3.12837]
  mu_s [0.791209, 0.637675, 0.559972, 0.538976]
  mu_d [0.767936, 0.637675, 0.502209, 0.538976]
  Kp [steer,rear] [9.24463, 0]; Kd [0.18092, 0]
  spawn roll 0.0903796 rad; rate -0.21792 rad/s
  gyro bias XYZ rad/s [-9.52816e-05, 1.26954e-05, 1.48451e-07]
  gravity bias YZ [-0.00200569, -0.00128185]
E3 [30.20, 60.00) s:
  mass x [1.00374, 0.883257, 1.08934, 0.883413]
  CoM XYZ mm [-1.56465, 1.33965, -4.50909]
  mu_s [0.71155, 0.791209, 0.640867, 0.640867]
  mu_d [0.71155, 0.767936, 0.556841, 0.556841]
  Kp [steer,rear] [9.87806, 0]; Kd [0.192029, 0]
  spawn roll 0.0277374 rad; rate -0.123222 rad/s
  gyro bias XYZ rad/s [7.64843e-05, 0.000117377, -9.60973e-05]
  gravity bias YZ [-0.002893, 0.00274328]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 61 · full 60 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 61; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 25.10) s:
  mass x [1.04593, 1.00667, 0.908218, 0.97081]
  CoM XYZ mm [-3.09935, -2.72854, 2.47374]
  mu_s [0.532645, 0.576972, 0.681711, 0.693205]
  mu_d [0.532645, 0.576972, 0.611756, 0.693205]
  Kp [steer,rear] [10.6263, 0]; Kd [0.202767, 0]
  spawn roll 0.0778073 rad; rate 0.198054 rad/s
  gyro bias XYZ rad/s [0.000370692, -0.000214087, 0.000211219]
  gravity bias YZ [0.00040144l, -0.00307435]
E2 [25.10, 55.10) s:
  mass x [0.987953, 1.03808, 0.810178, 1.14461]
  CoM XYZ mm [2.43007, -0.791136, -0.0987351]
  mu_s [0.787792, 0.533455, 0.785466, 0.592046]
  mu_d [0.617134, 0.533455, 0.52258, 0.534947]
  Kp [steer,rear] [9.74718, 0]; Kd [0.192989, 0]
  spawn roll 0.0754349 rad; rate 0.18036 rad/s
  gyro bias XYZ rad/s [-0.000316195, 0.000214386, -6.07899e-05]
  gravity bias YZ [0.00331178, -0.00432885]
E3 [55.10, 60.00) s:
  mass x [0.98811, 1.0164, 1.02224, 1.03655]
  CoM XYZ mm [-3.74888, 2.32713, 1.3504]
  mu_s [0.785466, 0.751971, 0.598343, 0.598343]
  mu_d [0.52258, 0.541224, 0.598343, 0.598343]
  Kp [steer,rear] [9.23459, 0]; Kd [0.208641, 0]
  spawn roll 0.0454457 rad; rate -0.126205 rad/s
  gyro bias XYZ rad/s [0.000118363, 0.000193584, 6.91042e-05]
  gravity bias YZ [-0.00215871, -0.000615227]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 61; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.24) s:
  mass x [0.925073, 0.889727, 0.872804, 1.14407]
  CoM XYZ mm [3.57542, -3.96438, -2.1091]
  mu_s [0.538976, 0.71008, 0.568073, 0.749678]
  mu_d [0.538976, 0.59371, 0.568073, 0.741429]
  Kp [steer,rear] [9.87736, 0]; Kd [0.209599, 0]
  spawn roll 0.0844232 rad; rate 0.0744198 rad/s
  gyro bias XYZ rad/s [0.000148847, -0.000162669, 0.000397757]
  gravity bias YZ [-4.9864e-05, -0.000578105]
E2 [0.24, 30.24) s:
  mass x [1.05433, 1.16247, 0.914729, 0.93478]
  CoM XYZ mm [-0.741662, 0.0209448, -0.15809]
  mu_s [0.732187, 0.723429, 0.748289, 0.736077]
  mu_d [0.631664, 0.723429, 0.689499, 0.699455]
  Kp [steer,rear] [9.65851, 0]; Kd [0.187563, 0]
  spawn roll 0.0733542 rad; rate 0.128861 rad/s
  gyro bias XYZ rad/s [-0.000183336, -0.000173914, 0.000224317]
  gravity bias YZ [0.00101996, -0.00164359]
E3 [30.24, 30.68) s:
  mass x [0.920135, 0.923472, 1.10074, 0.844097]
  CoM XYZ mm [-4.73483, 4.79922, -4.08565]
  mu_s [0.733515, 0.640867, 0.704749, 0.611239]
  mu_d [0.558392, 0.556841, 0.541926, 0.611239]
  Kp [steer,rear] [9.80817, 0]; Kd [0.210265, 0]
  spawn roll 0.0999161 rad; rate -0.15168 rad/s
  gyro bias XYZ rad/s [1.77661e-05, 0.000285011, -8.55378e-05]
  gravity bias YZ [0.000725614, -0.00029702]
E4 [30.68, 31.54) s:
  mass x [0.800301, 1.07805, 0.995308, 0.853685]
  CoM XYZ mm [-2.49548, -3.73241, -4.03448]
  mu_s [0.732187, 0.736077, 0.568073, 0.700521]
  mu_d [0.631664, 0.699455, 0.568073, 0.641329]
  Kp [steer,rear] [10.8423, 0]; Kd [0.206484, 0]
  spawn roll 0.0193762 rad; rate -0.0981116 rad/s
  gyro bias XYZ rad/s [-7.14101e-06, -1.94999e-05, -0.000164435]
  gravity bias YZ [0.00283172, 0.00119335]
E5 [31.54, 60.00) s:
  mass x [1.15984, 1.16839, 0.875318, 1.0121]
  CoM XYZ mm [-4.58364, -1.56581, 0.789727]
  mu_s [0.723429, 0.541939, 0.727426, 0.611239]
  mu_d [0.723429, 0.534359, 0.606358, 0.611239]
  Kp [steer,rear] [9.18896, 0]; Kd [0.196851, 0]
  spawn roll -0.0508898 rad; rate 0.101941 rad/s
  gyro bias XYZ rad/s [0.000356061, 1.52802e-05, -2.66949e-05]
  gravity bias YZ [-0.000579852, 2.96125e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 61; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.925073, 0.889727, 0.872804, 1.14407]
  CoM XYZ mm [3.57542, -3.96438, -2.1091]
  mu_s [0.538976, 0.71008, 0.568073, 0.749678]
  mu_d [0.538976, 0.59371, 0.568073, 0.741429]
  Kp [steer,rear] [9.87736, 0]; Kd [0.209599, 0]
  spawn roll 0.0844232 rad; rate 0.0744198 rad/s
  gyro bias XYZ rad/s [0.000148847, -0.000162669, 0.000397757]
  gravity bias YZ [-4.9864e-05, -0.000578105]
E2 [0.28, 11.36) s:
  mass x [0.84446, 1.05807, 1.1099, 0.900103]
  CoM XYZ mm [0.461385, -4.53228, -0.61191]
  mu_s [0.71155, 0.732187, 0.538434, 0.69316]
  mu_d [0.71155, 0.631664, 0.538434, 0.69316]
  Kp [steer,rear] [10.2815, 0]; Kd [0.21241, 0]
  spawn roll -0.0540162 rad; rate -0.00341233 rad/s
  gyro bias XYZ rad/s [5.52317e-05, 0.000117328, -0.000106188]
  gravity bias YZ [0.0036445, 0.00104761]
E3 [11.36, 11.76) s:
  mass x [0.976073, 0.952244, 1.14694, 0.947814]
  CoM XYZ mm [-4.80501, 2.48823, 3.20485]
  mu_s [0.736077, 0.732187, 0.69316, 0.640016]
  mu_d [0.699455, 0.631664, 0.69316, 0.513141]
  Kp [steer,rear] [10.6694, 0]; Kd [0.18178, 0]
  spawn roll 0.0879026 rad; rate 0.217445 rad/s
  gyro bias XYZ rad/s [1.17678e-05, -0.000241729, 8.43254e-05]
  gravity bias YZ [0.00272294, 0.00291104]
E4 [11.76, 36.40) s:
  mass x [1.00683, 1.04085, 0.836539, 0.859487]
  CoM XYZ mm [1.19432, -4.86366, -1.23454]
  mu_s [0.69316, 0.640867, 0.736077, 0.732187]
  mu_d [0.69316, 0.556841, 0.699455, 0.631664]
  Kp [steer,rear] [9.68612, 0]; Kd [0.214206, 0]
  spawn roll -0.00215132 rad; rate 0.102432 rad/s
  gyro bias XYZ rad/s [-0.000253889, 0.000169451, -2.6825e-05]
  gravity bias YZ [-0.00186643, 0.00197135]
E5 [36.40, 37.46) s:
  mass x [1.19775, 1.03569, 1.05299, 0.845648]
  CoM XYZ mm [-0.448106, 3.76588, 4.11845]
  mu_s [0.71155, 0.640016, 0.733515, 0.640016]
  mu_d [0.71155, 0.513141, 0.558392, 0.513141]
  Kp [steer,rear] [9.41897, 0]; Kd [0.201391, 0]
  spawn roll -0.0667015 rad; rate -0.0125078 rad/s
  gyro bias XYZ rad/s [0.000240755, -0.000445043, -0.000135583]
  gravity bias YZ [-0.000442733, -0.00151677]
E6 [37.46, 60.00) s:
  mass x [1.1405, 0.954225, 0.936967, 0.98647]
  CoM XYZ mm [3.66501, -3.95416, -4.22501]
  mu_s [0.611239, 0.748289, 0.749678, 0.637675]
  mu_d [0.611239, 0.689499, 0.741429, 0.637675]
  Kp [steer,rear] [9.36258, 0]; Kd [0.216975, 0]
  spawn roll 0.0713233 rad; rate -0.235914 rad/s
  gyro bias XYZ rad/s [-4.11503e-05, -0.000301463, 0.000178844]
  gravity bias YZ [-0.000992252, -0.00333554]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 62 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 62
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 62
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 62
DR · 0.00–60.00 s

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

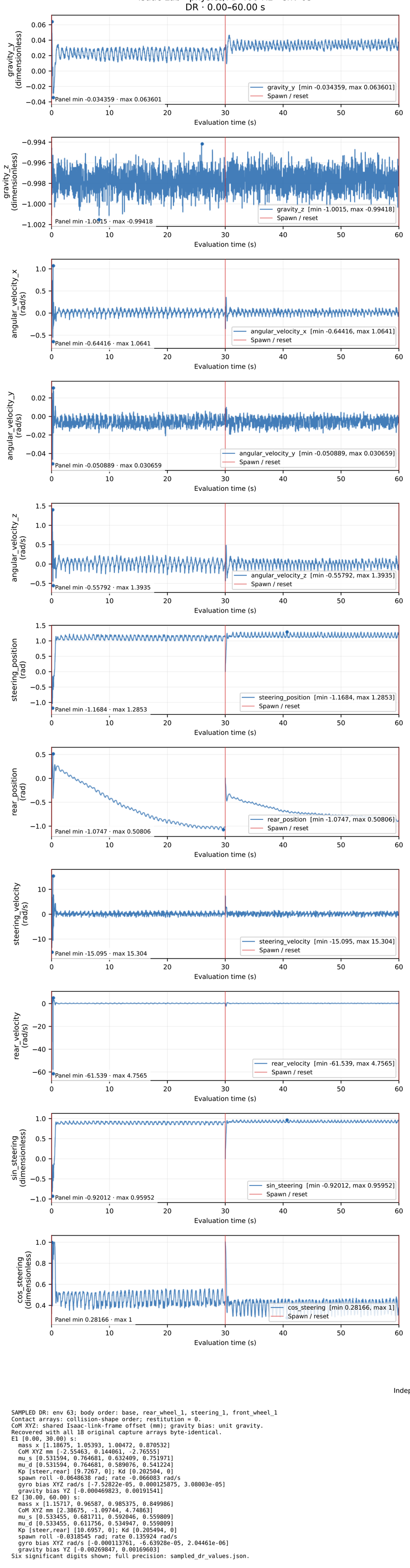
```
SAMPLED DR: env 62; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.01205, 0.910727, 1.1656, 1.1771]
  CoM XYZ mm [2.69511, 0.666026, -1.98537]
  mu_s [0.647391, 0.751971, 0.579974, 0.647391]
  mu_d [0.647391, 0.541224, 0.579974, 0.647391]
  Kp [steer,rear] [10.7399, 0]; Kd [0.186251, 0]
  spawn roll 0.00838556 rad; rate -0.0323896 rad/s
  gyro bias XYZ rad/s [0.000244845, -4.33467e-05, -9.10838e-05]
  gravity bias YZ [-0.00124117, -0.000246043]
E2 [30.00, 60.00) s:
  mass x [0.96663, 0.928456, 0.920087, 1.10597]
  CoM XYZ mm [0.830397, 2.0506, 0.429973]
  mu_s [0.531594, 0.794338, 0.532645, 0.774409]
  mu_d [0.531594, 0.526209, 0.532645, 0.561095]
  Kp [steer,rear] [9.85539, 0]; Kd [0.191222, 0]
  spawn roll 0.0110493 rad; rate -0.0630774 rad/s
  gyro bias XYZ rad/s [0.000312513, 4.54513e-05, -4.47451e-06]
  gravity bias YZ [0.000620301, -0.000547595]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 62; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.13543, 1.07686, 0.886436, 0.850165]
  CoM XYZ mm [-1.24931, 2.53889, 2.35188]
  mu_s [0.791209, 0.640016, 0.669571, 0.631146]
  mu_d [0.767936, 0.513141, 0.669571, 0.631146]
  Kp [steer,rear] [10.6484, 0]; Kd [0.201532, 0]
  spawn roll 0.059666 rad; rate 0.0300787 rad/s
  gyro bias XYZ rad/s [0.000176113, 0.000152281, -0.000194907]
  gravity bias YZ [0.00362988, 0.000554935]
E2 [0.42, 30.42) s:
  mass x [1.02572, 1.03521, 0.926834, 1.18918]
  CoM XYZ mm [-4.81969, -4.60651, 1.93796]
  mu_s [0.541939, 0.559972, 0.669571, 0.736077]
  mu_d [0.534359, 0.502209, 0.669571, 0.699455]
  Kp [steer,rear] [10.4723, 0]; Kd [0.205074, 0]
  spawn roll 0.0246067 rad; rate 0.244275 rad/s
  gyro bias XYZ rad/s [-0.000401951, 0.000148931, 6.53948e-06]
  gravity bias YZ [0.00162322, 0.00193128]
E3 [30.42, 30.84) s:
  mass x [1.10759, 1.09162, 0.873657, 1.17475]
  CoM XYZ mm [0.415086, -2.26386, -0.216222]
  mu_s [0.633024, 0.616244, 0.568073, 0.69316]
  mu_d [0.568172, 0.586498, 0.568073, 0.69316]
  Kp [steer,rear] [9.83548, 0]; Kd [0.215189, 0]
  spawn roll -0.0990009 rad; rate -0.0385385 rad/s
  gyro bias XYZ rad/s [-4.38903e-05, -5.06665e-06, 0.000156636]
  gravity bias YZ [-0.00315005, -0.000378556]
E4 [30.84, 31.28) s:
  mass x [1.01935, 0.854325, 0.956056, 0.900375]
  CoM XYZ mm [-4.14943, 1.34356, 3.07301]
  mu_s [0.748289, 0.538434, 0.631146, 0.637675]
  mu_d [0.689499, 0.538434, 0.631146, 0.637675]
  Kp [steer,rear] [9.42288, 0]; Kd [0.186403, 0]
  spawn roll 0.0655409 rad; rate 0.0403173 rad/s
  gyro bias XYZ rad/s [-0.000120343, -9.64318e-05, 0.000125775]
  gravity bias YZ [-0.00197305, -0.00103767]
E5 [31.28, 60.00) s:
  mass x [0.850267, 1.19756, 1.19215, 0.985976]
  CoM XYZ mm [0.438962, -4.99748, 3.52757]
  mu_s [0.71155, 0.541939, 0.733515, 0.69316]
  mu_d [0.71155, 0.534359, 0.558392, 0.69316]
  Kp [steer,rear] [10.2673, 0]; Kd [0.197007, 0]
  spawn roll -0.0176868 rad; rate 0.159026 rad/s
  gyro bias XYZ rad/s [-5.37826e-05, 2.1177e-05, -0.000322962]
  gravity bias YZ [-0.00347019, 0.000245845]
Six significant digits shown; full precision: sampled_dr_values.json.
```

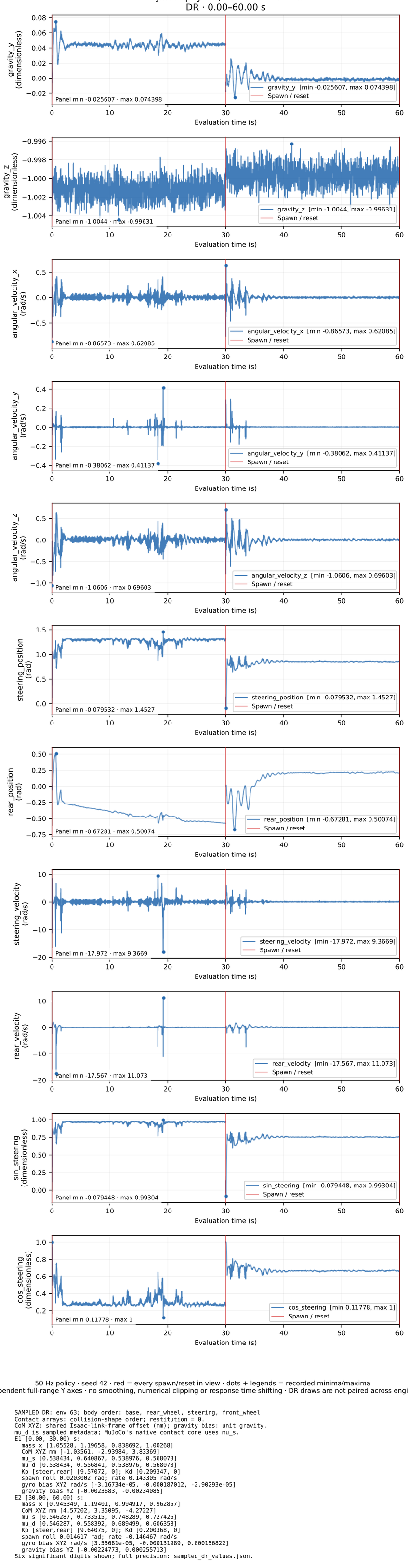
```
SAMPLED DR: env 62; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.13543, 1.07686, 0.886436, 0.850165]
  CoM XYZ mm [-1.24931, 2.53889, 2.35188]
  mu_s [0.791209, 0.640016, 0.669571, 0.631146]
  mu_d [0.767936, 0.513141, 0.669571, 0.631146]
  Kp [steer,rear] [10.6484, 0]; Kd [0.201532, 0]
  spawn roll 0.059666 rad; rate 0.0300787 rad/s
  gyro bias XYZ rad/s [0.000176113, 0.000152281, -0.000194907]
  gravity bias YZ [0.00362988, 0.000554935]
E2 [30.00, 30.34) s:
  mass x [1.10435, 0.832656, 1.02369, 0.801257]
  CoM XYZ mm [2.57136, -3.06342, -1.87806]
  mu_s [0.71008, 0.633024, 0.704749, 0.538434]
  mu_d [0.59371, 0.568172, 0.541926, 0.538434]
  Kp [steer,rear] [9.83748, 0]; Kd [0.197791, 0]
  spawn roll 0.0647167 rad; rate 0.14349 rad/s
  gyro bias XYZ rad/s [-7.07498e-06, 0.000204651, -0.000314145]
  gravity bias YZ [-0.00081528, -0.00145384]
E3 [30.34, 30.76) s:
  mass x [0.84416, 1.00113, 1.17612, 0.844773]
  CoM XYZ mm [0.435414, 1.86383, -0.135084]
  mu_s [0.732187, 0.538976, 0.749678, 0.538976]
  mu_d [0.631664, 0.538976, 0.741429, 0.538976]
  Kp [steer,rear] [9.90804, 0]; Kd [0.198313, 0]
  spawn roll -0.0646306 rad; rate -0.116687 rad/s
  gyro bias XYZ rad/s [0.000238552, 0.000172162, 0.000378569]
  gravity bias YZ [-0.00113772, -0.00426276]
E4 [30.76, 60.00) s:
  mass x [0.843686, 0.984973, 0.877347, 0.823193]
  CoM XYZ mm [-3.60021, 4.60623, 0.829765]
  mu_s [0.69316, 0.546287, 0.791209, 0.546287]
  mu_d [0.69316, 0.546287, 0.767936, 0.546287]
  Kp [steer,rear] [10.8657, 0]; Kd [0.214676, 0]
  spawn roll 0.0416617 rad; rate 0.0262887 rad/s
  gyro bias XYZ rad/s [0.000278631, 0.00010735, -0.000126949]
  gravity bias YZ [0.00130734, 0.00124375]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 63 · full 60 s

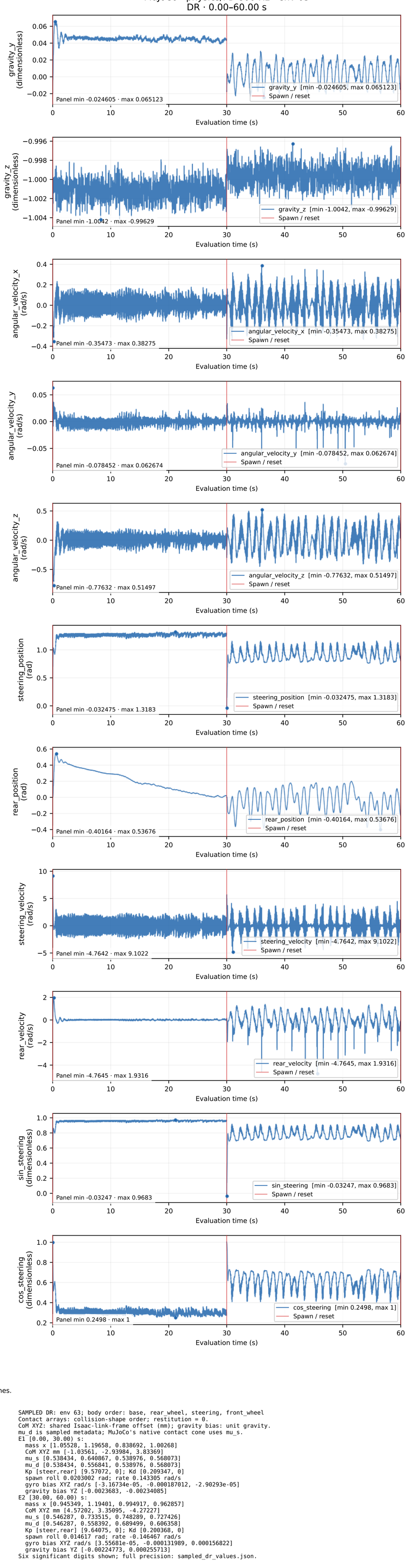
Isaac Lab · physics/PD 400 Hz · env 63
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 63
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 63
DR · 0.00–60.00 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 63; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.18675, 1.05393, 1.00472, 0.870532]
  CoM XYZ mm [-2.55463, 0.144061, -2.76555]
  mu_s [0.531594, 0.764681, 0.632409, 0.751971]
  mu_d [0.531594, 0.764681, 0.589076, 0.541224]
  Kp [steer,rear] [9.7267, 0]; Kd [0.202504, 0]
  spawn roll -0.0648638 rad; rate -0.066083 rad/s
  gyro bias XYZ rad/s [-7.52822e-05, 0.000125875, 3.08003e-05]
  gravity bias YZ [-0.000469823, 0.00191541]
E2 [30.00, 60.00) s:
  mass x [1.15717, 0.96587, 0.985375, 0.849986]
  CoM XYZ mm [2.38675, -1.09744, 4.74863]
  mu_s [0.533455, 0.681711, 0.592046, 0.559809]
  mu_d [0.533455, 0.611756, 0.534947, 0.559809]
  Kp [steer,rear] [10.6957, 0]; Kd [0.205494, 0]
  spawn roll -0.0318545 rad; rate 0.135924 rad/s
  gyro bias XYZ rad/s [-0.000113761, -6.63928e-05, 2.04461e-06]
  gravity bias YZ [-0.00269847, 0.00169603]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 63; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05528, 1.19658, 0.838692, 1.00268]
  CoM XYZ mm [-1.03561, -2.93984, 3.83369]
  mu_s [0.538434, 0.640867, 0.538976, 0.568073]
  mu_d [0.538434, 0.556841, 0.538976, 0.568073]
  Kp [steer,rear] [9.57072, 0]; Kd [0.209347, 0]
  spawn roll 0.0203002 rad; rate 0.143305 rad/s
  gyro bias XYZ rad/s [-3.16734e-05, -0.000187012, -2.90293e-05]
  gravity bias YZ [-0.0023683, -0.00234085]
E2 [30.00, 60.00) s:
  mass x [0.945349, 1.19401, 0.994917, 0.962857]
  CoM XYZ mm [4.57202, 3.35095, -4.27227]
  mu_s [0.546287, 0.733515, 0.748289, 0.727426]
  mu_d [0.546287, 0.558392, 0.689499, 0.606358]
  Kp [steer,rear] [9.64075, 0]; Kd [0.200368, 0]
  spawn roll 0.014617 rad; rate -0.146467 rad/s
  gyro bias XYZ rad/s [3.55681e-05, -0.000131989, 0.000156822]
  gravity bias YZ [-0.00224773, 0.000255713]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 63; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05528, 1.19658, 0.838692, 1.00268]
  CoM XYZ mm [-1.03561, -2.93984, 3.83369]
  mu_s [0.538434, 0.640867, 0.538976, 0.568073]
  mu_d [0.538434, 0.556841, 0.538976, 0.568073]
  Kp [steer,rear] [9.57072, 0]; Kd [0.209347, 0]
  spawn roll 0.0203002 rad; rate 0.143305 rad/s
  gyro bias XYZ rad/s [-3.16734e-05, -0.000187012, -2.90293e-05]
  gravity bias YZ [-0.0023683, -0.00234085]
E2 [30.00, 60.00) s:
  mass x [0.945349, 1.19401, 0.994917, 0.962857]
  CoM XYZ mm [4.57202, 3.35095, -4.27227]
  mu_s [0.546287, 0.733515, 0.748289, 0.727426]
  mu_d [0.546287, 0.558392, 0.689499, 0.606358]
  Kp [steer,rear] [9.64075, 0]; Kd [0.200368, 0]
  spawn roll 0.014617 rad; rate -0.146467 rad/s
  gyro bias XYZ rad/s [3.55681e-05, -0.000131989, 0.000156822]
  gravity bias YZ [-0.00224773, 0.000255713]
Six significant digits shown; full precision: sampled_dr_values.json.
```